

Public Economics and Development

Week 11: Universal Income and Means-tested Welfare Programs Across the Globe

Seung-hun Lee

Taipei School of Economics
National Tsing Hua University

May 5th, 2026

UBI and Targeted Transfers: Where in the World?

Universal Basic Income (UBI)

- Kenya (GiveDirectly, 2017–): 20,000 recipients, 12-year horizon
- Finland (2017–18): 2,000 unemployed adults, 560 Euro/month
- Alaska Permanent Fund (universal dividend since 1982, \$1,000 - 2,000/year)
- Ongoing pilots in 20+ countries as of 2024

Means-tested / Targeted Programs

- 149 countries operate at least one social protection program
- CCTs in 66+ LMICs; cover ~760 million people
- Social safety nets account for 1.5% of GDP on average in LMICs
- Sub-Saharan Africa & South Asia: lowest coverage (<20% of poorest quintile)

UBI vs. Targeted Transfers: Key Debates

Case for UBI

- Eliminate exclusion errors
- Reduces stigma and administrative costs of targeting
- Provides a permanent income floor and insurance against shocks
- Evidence from pilots: no significant negative labor supply effects

Case for Targeted Transfers

- Concentrating resources on the poor raises impact per dollar
- CCTs generate human capital gains alongside income
- UBI at adequate levels is fiscally unaffordable for most LMICs
- Political palatable for broader public

Central tension

- **Universality** (coverage, low admin cost) vs. **Targeting** (sustainability, favor eligible poor)
- Neither dominates: optimal mix depends on fiscal/political constraint and objectives

Ongoing NIT experiment: Seoul Stepping Stone Income Project

Fun fact: I am part of the research team assessing this program

Who is eligible?

- Seoul residents with assets $\leq$ 326 million KRW ($\approx$ 236,000 USD)
- Cohort 1: income $\leq$ 50% of standard median income; Cohort 2: $\leq$ 85%
- Coverage $\uparrow$, exclusion errors $\downarrow$, and generosity $\uparrow$ than existing programs

Transfers varies across income: Poorer households receive more

- Amount: half of the shortfall between current earnings and the income threshold
- **50% phase-out rate** vs. 70% under existing program $\Rightarrow$ weaker labor disincentive

Experimental timeline: Assignment randomized among eligible applicants

- Cohort 1 (484 treated, 1,039 control): treatment runs Jul 2022 – Jun 2025
- Cohort 2 (1,100 treated, 2,488 control): treatment runs Jul 2023 – Jun 2025
- Survey data collected twice per year; administrative data annually

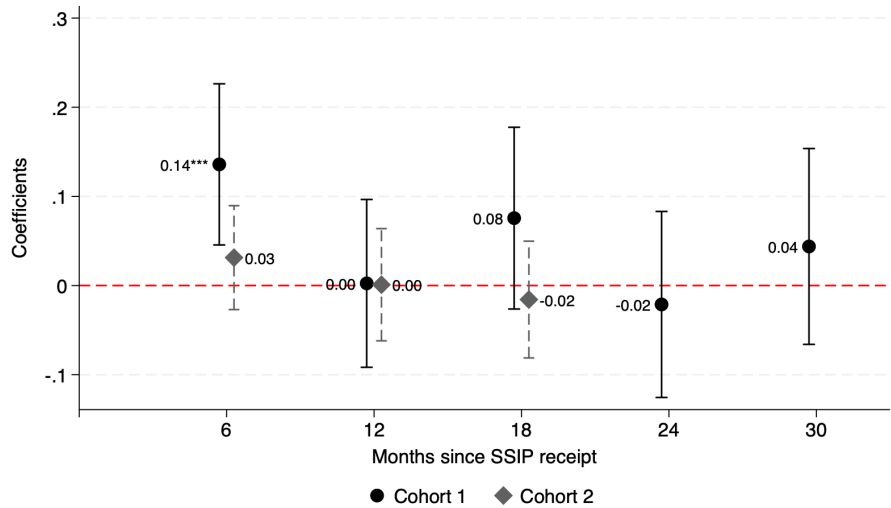
Increases in spending on essential and investment goods

	Expenditures					
	Total	Food	Utilities	Edu. & Train	Medical	Other
	(1)	(2)	(3)	(4)	(5)	(6)
SSIP x Post	10.960*** (3.378)	3.633*** (1.329)	0.758** (0.318)	0.620** (0.288)	1.502* (0.841)	4.445* (2.470)
Control Mean	198.5	54.03	13.50	1.043	10.86	119.1
Observations	20302	20302	20302	20302	20302	20302
R-Squared	0.441	0.337	0.203	0.021	0.060	0.378
Baseline Characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Strata Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
Wave Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes
Cohort Fixed Effect	Yes	Yes	Yes	Yes	Yes	Yes

Lower labor supply amongst recipients

	labor+busi income>0 (1)	Employed (2)	Worked Last Week (3)	Hours Worked (4)
SSIP x Post	-0.113*** (0.017)	-0.093*** (0.016)	-0.102*** (0.017)	-0.531 (0.728)
Control Mean	0.651	0.589	0.564	33.26
Observations	15998	20302	20302	11973
R-Squared	0.179	0.166	0.160	0.136
Baseline Char.	Yes	Yes	Yes	Yes
Strata Fixed Effect	Yes	Yes	Yes	Yes
Wave Fixed Effect	Yes	Yes	Yes	Yes
Cohort Fixed Effect	Yes	Yes	Yes	Yes

Short-lived mental health effects amongst recipients



Roadmap for today

1. Overview setup of the topic
2. Theoretical motivation across different types of welfare programs
 - 2.1 Moffit (2003): The Negative Income Tax and the Evolution of the US Welfare Policy
3. Studies on Universal Basic Income
 - 3.1 Hanna, Olken (2018): Universal Basic Income versus Targeted Transfers: Anti-Poverty Programs in Developing Countries
 - 3.2 Banerjee, Niehaus, Suri (2019): Universal Basic Income in the Developing World
4. Targeted transfers: A Look into Earned Income Tax Credit Program
 - 4.1 Eissa, Liebman (1996): Labor Supply response to Earned Income Tax Credit
 - 4.2 Kleven (2024): The EITC and the Extensive Margin: A Reappraisal

Moffit (2003):
The Negative Income Tax and the Evolution of
the US Welfare Policy

Q: How does negative income tax affect labor supply ?

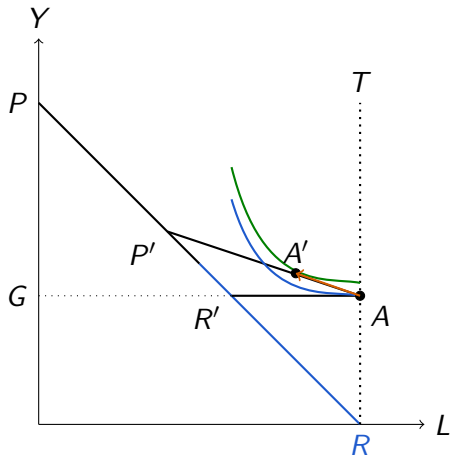
Debate over labor supply effects across different welfare programs

- Positive income tax: Portion of income claimed as taxes and increase in income
- Negative income tax (NIT): Send money back, but at decreasing rate as income rises
 - ▶ Provide certain level of benefits event at zero income
 - ▶ But size of benefits reduced with more income
- Early welfare designs had less interest in disincentives from welfare
 - ▶ Generally benefits reduced by \$1 for each \$1 of earned income
 - ▶ NIT: benefits reduced by $< \$1$ per \$1 of earned income

This paper summarizes key components of NIT and analyzes its impact on labor supply

- Labor supply effect of NIT vs others
- Provides studies on how some principles of NIT are adopted

Friedman's version of NIT



Househod budget

- Earned income Y + benefit B
- $Y = w(T - L) = wl$
- $B = \max\{0, G - tY\}$, where t is tax on Y

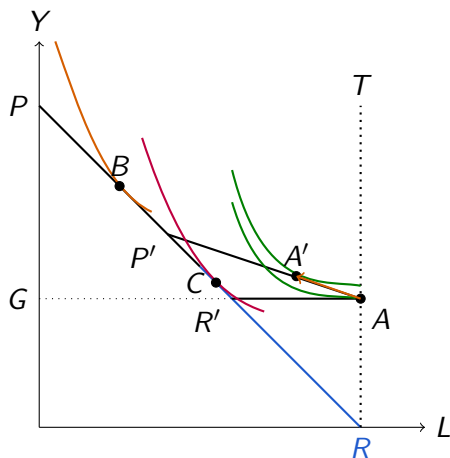
Minimum guaranteed income, with 100% tax

- Guarantees G
- Additional income fully taxed
- Likely to choose no work (point A)

NIT with lower than 100% tax rate

- Labor supply $\uparrow$ compared to above case
- Targeting efficiency solely based on income (Friedman 1962)

Ambiguous labor supply effect?



Different labor supply responses

- Possible to increase Y and L ($C \rightarrow AP'$)
- Possible to increase L for lower Y ($B \rightarrow AP'$)
- Total change in labor supply is ambiguous

Other noticeable effects

- Cost effective targeting? Not always
- More coverage cost with less labor possible if B and C re-optimize

Empiric evidence

- Labor $\downarrow$ in Seattle and Seoul ($t = 0.7, 0.5$)
 - ▶ Except for single mothers (Seattle)
 - ▶ And those on previous welfare programs with higher t (Seoul)+

Work requirements and categorical transfer systems

Work requirements definition

- Categorize population to those who can and cannot work
- Former receive benefits only if they satisfy minimum work hours (otherwise denied)
- Latter given some guaranteed income, that phases out with earned income

Work requirement is at odds with NIT

- Against minimal government: need welfare bureaucracy to administer the program
- Possibility to manipulate (mimick) into disability (where NIT purely based on income)

Proponents of work requirement

- Possible screening: Identify needy individuals who potentially benefit more
- Work requirement as 'ordeals' (Nichols, Zeckhause 1982)
- Tagging and additional costs help identify the needy when true measure is unavailable

Development of welfare in US: Welfare tax rates

Earned income tax credit (since 1996)

- Subsidy to earnings up to some point, which is phased out
- Lower minimum guaranteed income
- Also lower marginal tax rate on earned income (between 50-75%)
- Extended application of NIT: Wide range of t and eligibility based on income
- Heterogenous labor supply effect: Single mothers $\uparrow$, secondary earner work $\downarrow$

Other developments: Temporary assistance for needy families (TANF)

- Involved withdrawal of benefits with additional income at varying rates
- But also included components related to work requirements

Development of welfare in US: Others

History of work requirement programs in US

- Growing element since 1970s: After NIT proposal by Nixon Admin was defeated
- Minimum work hour started at 20 hrs per week (expanded to 30 hours)
- Exemption on work requirement vary by state
- Expenditure per recipient rose: Cost of regulating individual lives of welfare recipients
- Did screen out those who have less need for support, reducing caseloads
- Thus, room to create new programs to address different problems

Various means-tested programs

- Medicaid, Food stamps, Foster care, Supplemental Security Income etc
- Screening based on non-income, ordeal-like measures

Key messages

Takeaways

- Theoretical benefits of NIT not realized in some cases
 - ▶ Ambiguous labor supply effect
 - ▶ Gains from non-monetary screening mechanisms
- Some developments are consistent with NIT: EITC, for instance
- Others are not: Work requirement, means-tested programs
- Still, NIT serves as theoretical framework to compare against other existing programs
- Brought trade-offs between work incentives and support up front
 - ▶ Also for marriage, education, child-bearing, among outcomes
- Useful contribution to economic reasoning at public policy analysis

Hanna, Olken (2018):
Universal Basic Income versus Targeted
Transfers: Anti-Poverty Programs in Developing
Countries

Q: How do UBI and targeted programs address poverty?

Eradicating poverty worldwide is an important goal

- One of the SDGs articulated by the United Nations
- While progress has been made, there are caveats
 - ▶ Heavily driven by economic growth in India and China
 - ▶ Even with future growth, still require substantial reduction in poverty
- National-level transfers have an important role in this context
 - ▶ Some based on targeting: Proxy means test and community based test
 - ▶ Alternative is to reduce administrative costs with UBI

This paper compares trade-offs observed across targeting and UBI

- Examine how both programs are conducted in developing countries
- Investigates trade-offs based on evidence from Indonesia and Peru
- Then draw implications on broader policy debate on redistribution

Relationship between UBI and Income taxation problem

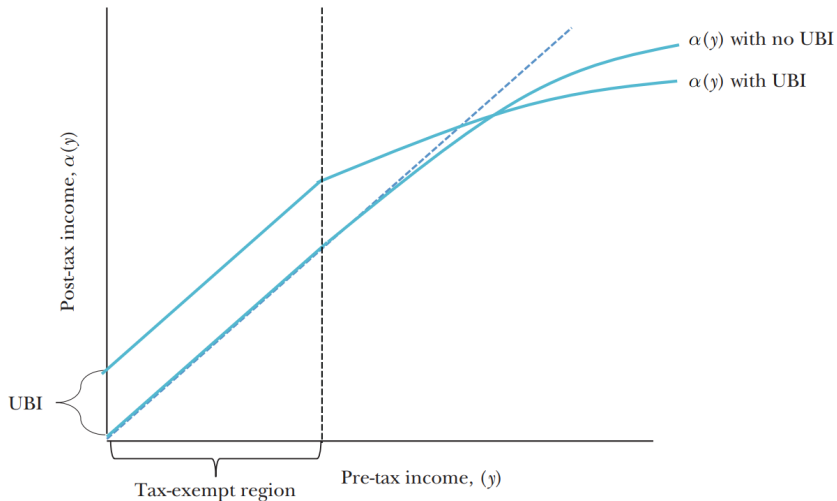
Design of UBI and taxation

- UBIs ultimately need to be financed via domestic taxation
 - ▶ In general, external sources such as ODA takes up a very small share of the budget
- Suppose UBI is given to all households
- Households with income $y > 0$ pay taxes progressive to their income to finance UBI

Trade-offs involved with UBI in developing countries

- Many individuals (80% in some case) have reported income below tax thresholds
 - ▶ Sometimes due to actually being poor, but also due to informal economies
- Need to finance UBI from small group: High marginal tax rate for the few
- Fewer overall funds + Large labor disincentives on productive population

Tax schedule with and without UBI, including tax exemption



Targeting in developing countries

How targeting works when income is not perfectly observable

- Proxy means test based on census
 - ▶ Door-to-door visits to examine visible assets and personal information
 - ▶ Rarely on self-reported basis (but people can misreport)
- Use the information to predict household income
 - ▶ Combine with other dataset collected for different purposes
 - ▶ Regress poverty from other dataset with assets collected through census
- Because of the *predicted* nature, inclusion and exclusion errors exist

Other methods of targeting

- Community-based measures: Eligibility determined by community members
 - ▶ Use localized information: Fill in gaps of standard measure but difficult to cross-compare
- Conditional transfers: Set conditions on receipt of benefits
 - ▶ Incentive effects (more education) targeting effects (may exclude very poor)

Comparison of UBI and targeted transfers in Indonesia and Peru

Targeting program included in the study

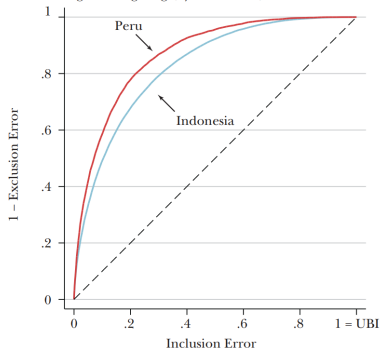
- Peru: CCT for poor families (Juntos)
- Indonesia: Nationwide UCT for poor household (BLT)
- Simulate targeting with assets from household data (2010-11) and varying thresholds

Key findings on performance

- Exclusion/inclusion error trade-off:
 - ▶ To reach 80% of target beneficiaries, need to tolerate 22-31% inclusion error
- Narrow vs wide targets on fixed budget
 - ▶ Transfers to per household fall as more people are allowed in
- Welfare comparison
 - ▶ Narrowly targeted programs perform better in terms of welfare
 - ▶ Large benefit-per-capita to the poorest of the poor
 - ▶ This holds even with large inclusion errors, but horizontal equity concerns may follow

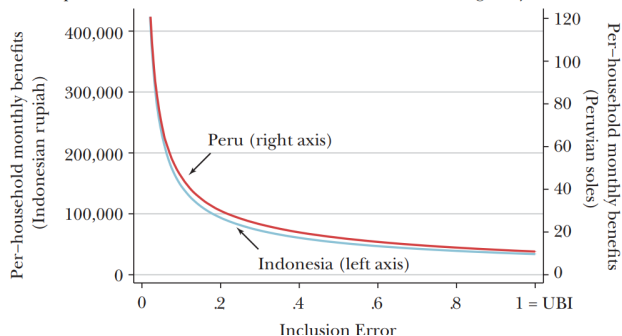
Error tradeoff and per household benefits

A: ROC Curves for Program Targeting (by Household)



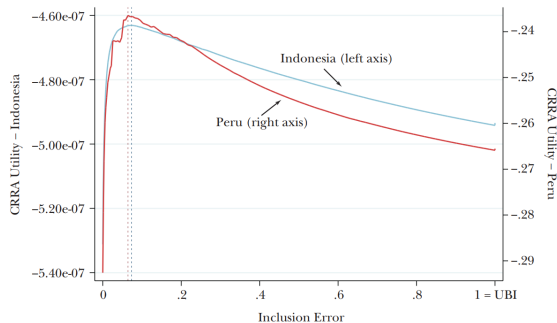
(a) Inclusion-exclusion error trade-off

B: Benefits per Household versus Inclusion Error for Different Eligibility Cutoffs

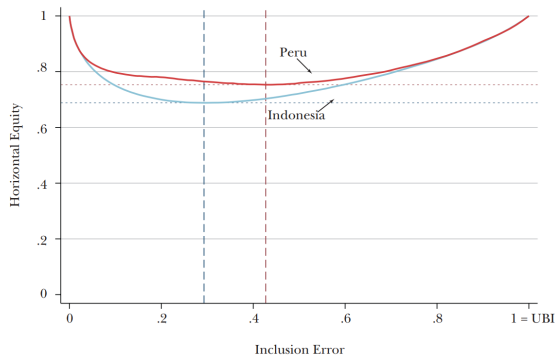


(b) Welfare amount per household

Welfare and horizontal equity



(a) Social welfare across different simulations



(b) Horizontal equity across different simulations

Banerjee, Niehaus, Suri (2019): Universal Basic Income in the Developing World

Q: How does UBI affect individuals in developing countries?

Raising income of the poor a pivotal goal in development economics and policy

- Universal basic income achieves this goal by design
- Many funders and taxpayers worry about simply giving away money
 - ▶ Worry about being too reliant on these transfers and/or wasting it away
- More cost-effective way to raise income?
 - ▶ Missing (credit) market, psychological burden of poverty more pressing and long-run
 - ▶ Large-scale infrastructural investment?
 - ▶ Perhaps a targeted income support does better, even in developing countries

This paper explores three questions on the effects of universal basic income

- What does individuals do with incremental income?
- Would UBI unlock further economic growth?
- Is it wise to give money to everyone, as opposed to targeting?

Where does incremental income go?

Where does the money for UBI come from?

- Repurposed foreign aid, re-allocating existing tax revenue, creating new taxes
- Need to persuade many stakeholders
 - ▶ Technocrats who design the policy
 - ▶ Tax payer who are also potential voters
- Even a *perceived* waste of UBIs (e.g. alcohol, drugs) reduces support for UBI

What do we know so far?

- UBI receipt reduced spending on temptation goods (alcohol, smoking etc)
- Various positive effects on income, assets, food consumption, schooling, health etc
 - ▶ These results, however, vary greatly across contexts
 - ▶ May reflect the feature that recipients have great autonomy with cash transfers
 - ▶ Imply that UBI may be suited for *collective sets of* goals as opposed to *particular* goal

What we do not know about where money goes

Full-scale nationwide coverage is rare

- Coverage of the programs in developing countries is not 100%
- Broad coverage could induce general equilibrium effects
 - ▶ Especially if there is a meaningful interaction between own and neighbor's income
 - ▶ If more in a given region is treated, could potentially affect market price and wages

Transfers to household vs transfers to each individual in the household

- Status quo: Typically goes to household head or mothers
- Individualizing transfers may change who decides how money is spent

No clear evidence of long-run effects of UBI

- Long-run UBI = Continuous inflow of income
- Savings may decrease ($\because$ less individual needed to smooth consumption)
- Increase tolerance for risk-taking

Does UBI relax constraints placed on economic growth?

Financial constraints faced by the poor

- No credit market: Cannot save or borrow to make future investments
- Underdeveloped insurance market: Shy away from making investments
- UBI may not necessarily be the best tool to address these, but still useful
 - ▶ But we do not exactly know which constraints bind for whom (and cannot target)
 - ▶ UBI, then, may serve as a catchall pro-growth policy in this context

What we find from various microcredit experiments and grant provisions

- Microcredit: Small loans to impoverished entrepreneurs w/o access to formal banks
- On average, borrowers do not use loans to grow businesses
- Heterogeneities: Those who already have business increase earnings (vs late starters)
 - ▶ For those facing credit constraint, relaxing it through income provision helps
 - ▶ But not everyone faces credit constraint: UBI alone does not relax these

Other constraints faced in developing countries

Lack of insurance markets in developing countries

- Evidence of more, high-risk & high-return investments following subsidized insurance
- Particularly among the more educated, and relatively wealthier population
- Can UBI offset constraints from uninsured risk?
 - ▶ Could potentially help meet basic needs, which is required prior to taking risks
 - ▶ Very little empirical research

Constraints from psychological factors

- Poverty leads to costs in mental bandwidth
- Graduation programs with training increase earnings/assets through aspirations
 - ▶ But limited impacts on those who had very little to begin with
- UBI could relax mental toll by reducing worries about making ends meet
 - ▶ Positive effects on mental health, decreases in suicides
 - ▶ But long-run effects not yet consolidated

Better to target or not?

Measuring poverty in developing countries

- Mostly on household poverty, as opposed to individual poverty
- Inclusion/exclusion errors: What if poorest individuals are not from poorest hhs?
- Costs of targeting: High especially in presence of informal economies
- Difficult to update poverty measure: Hard to catch changes in distribution of poverty
- Requires functioning bureaucracies: Otherwise opportunities for corruption

Can UBI schemes do better? Above problems may also exist here

- Still need universal identification system to prevent double counting and exclusion
 - ▶ Incurs large one-time investment
- Still face problem of accurately measuring poverty
- Also require capable bureaucracies to implement

Key messages from the paper

Studies on the UBI (and other targeting programs) are in evolving phase

- Many effects are very heterogeneous across different groups of individuals
- This makes generalizing the effects of these programs difficult
- So far, UBI does have positive measure across many measures
- But if the fix is targeting a *particular* measure, UBI may not be first-best
 - ▶ Health, education investments, consumption, lumpy investments
 - ▶ But if targeting a particular problem, better to design programs specific for that purpose
- So is it better to switch existing programs to a single transfer?
 - ▶ If so, people will buy these services on the market with the given money
 - ▶ But private sector (markets) for these goods and services do not exist in many places
 - ▶ Thus, this paper is NOT about advocating switch from existing programs to cash transfers

Eissa, Liebman (1996):
Labor Supply response to Earned Income Tax
Credit

Q: How does expanding EITC affect labor supply?

Traditional welfare for children reduced benefits 1-for-1 with additional income

- Theory and evidence point to decline in labor force participation
- Reforms in the US around 1990s refunds credit in excess of tax liability
 - ▶ Thus, welfare benefits decrease less than \$1 for each \$ of additional income
- There are those who predict that EITC will encourage labor supply
- But this is contentious: Many recipients are in phase-out or constant credit region

This paper documents classical evidence of labor supply effects of expanded EITC

- Leverages a major welfare reform in 1986 (TRA86)
- Comparing single women with children to single women with no children
- Investigates changes in labor force participation and hours worked

Eligibility and identification

Who is eligible and how much are they credited?

- Nonzero earned income (labor, business income)
- Adjusted gross income and earned income below specific amount (\$28,495 in 1996)
- Must have children, grandchildren, stepchild, or foster child below age 19 or disabled
- Credit and liability for households with children of 2
 - ▶ Phase in rate of 40% for first \$8,890 (max credit \$3,356), constant until \$ 11,610
 - ▶ Phase out from \$11,610 at 21.06% rate - no longer eligible at \$28,495

Identification uses DID with the expansion of EITC in 1987 (passed in 1986)

- Rise in phase-in credit rate, maximum credit, reduction in phase-out rate
- Also came in with other increased deductions that reduced tax burdens
- Single women with children (treated) vs single women without children (control)
 - ▶ Largest group eligible for EITC vs similar characteristics but unaffected by treatment
 - ▶ No need to worry about joint labor supply decisions

Increased labor force participation among single women w/ children

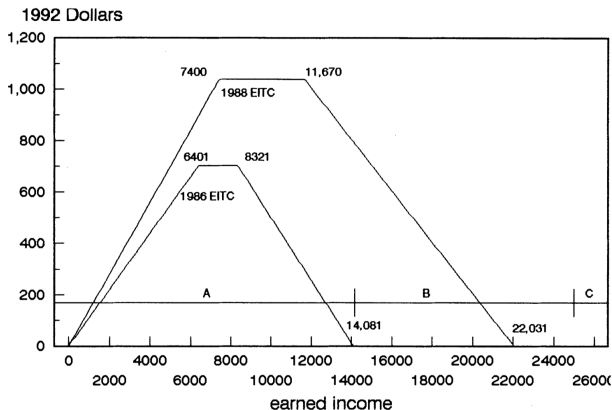
	Pre-TRA86 (1)	Post-TRA86 (2)	Difference (3)	Difference-in- differences (4)
<i>A. Treatment group:</i>				
With children [20,810]	0.729 (0.004)	0.753 (0.004)	0.024 (0.006)	
<i>Control group:</i>				
Without children [46,287]	0.952 (0.001)	0.952 (0.001)	0.000 (0.002)	0.024 (0.006)
<i>B. Treatment group:</i>				
Less than high school, with children [5396]	0.479 (0.010)	0.497 (0.010)	0.018 (0.014)	
<i>Control group 1:</i>				
Less than high school, without children [3958]	0.784 (0.010)	0.761 (0.009)	-0.023 (0.013)	0.041 (0.019)
<i>Control group 2:</i>				
Beyond high school, with children [5712]	0.911 (0.005)	0.920 (0.005)	0.009 (0.007)	0.009 (0.015)
<i>C. Treatment group:</i>				
High school, with children [9702]	0.764 (0.006)	0.787 (0.006)	0.023 (0.008)	
<i>Control group 1:</i>				
High school, without children [16,527]	0.945 (0.002)	0.943 (0.003)	-0.002 (0.004)	0.025 (0.009)
<i>Control group 2:</i>				
Beyond high school, with children [5712]	0.911 (0.005)	0.920 (0.005)	0.009 (0.007)	0.014 (0.011)

Also confirmed through probit regression

$$\Pr(\text{Ifp}_{it} = 1) = \Phi(\alpha + \gamma_0 \text{Kids}_i + \gamma_1 \text{Post86}_t + \gamma_3 \text{Kids}_i \times \text{Post86}_t + \beta Z_{it})$$

Variables	Sample: all unmarried women (Column titles indicate covariates controlled)					
	Without covariates (1)	Demographic characteristics (2)	Unemployment and AFDC (3)	State dummies (4)	Second child dummy (5)	Separate year interactions (6)
<i>Coefficient estimates</i>						
Kids (γ_0)	-1.053 (.020)	-0.250 (.029)	-1.403 (.106)	-1.438 (.108)	-1.458 (.110)	-1.462 (.110)
Post86 (γ_1)	-0.001 (.028)	0.019 (.031)	-0.152 (.067)	-0.104 (.069)	-0.094 (.069)	—
Kids $\times$ Post86 (γ_2)	0.069 (.027)	0.074 (.030)	0.103 (.037)	0.113 (.037)	0.087 (.043)	—
Kids $\times$ 1988	—	—	—	—	—	0.033 (.057)
Kids $\times$ 1989	—	—	—	—	—	0.116 (.058)
Kids $\times$ 1990	—	—	—	—	—	0.112 (.057)
Second child $\times$ post86	—	—	—	—	0.051 (.043)	—
<i>Predicted participation response for treatment group</i>						
	—	.019 (.008)	.026 (.010)	.028 (.009)	.022 (.009)	.008 (.014)
						.008 (.015)
						.029 (.015)

Expected effects on working hours



Region A ($0 \leq y \leq 14,081$): Ambiguous

- Income and substitution mixed
- Strength of effect depends on y

Region B ($14,081 \leq y \leq 22,031$): Hours $\downarrow$

- Marginal tax rate per additional y $\uparrow$
- Some just above max credit may reduce hours (and y) to receive EITC

Region C: Likely unaffected

- High-earners
- May not be worth it to reduce y

Limited working hour changes among those already working

Dependent variable:	Annual hours	Annual hours	Annual hours	Annual hours	Annual weeks	Annual weeks
Variables	All single women with hours > 0 (1)	Less than high school with hours > 0 (2)	All single women (3)	Less than high school (4)	All single women with hours > 0 (5)	All single women (6)
Coefficient estimates						
Other income (1000s)	-21.83 (.61)	-26.81 (2.93)	-29.92 (.62)	-56.65 (2.46)	-0.433 (.012)	-0.670 (.014)
Number of preschool children	-66.28 (10.42)	-72.21 (25.57)	-136.49 (9.18)	-107.94 (16.92)	-1.833 (.214)	-3.944 (.207)
Nonwhite	-140.94 (11.77)	-142.84 (41.29)	-209.80 (12.43)	-266.32 (36.14)	-2.680 (.241)	-4.788 (.281)
Age	786.82 (22.38)	475.01 (64.29)	576.16 (23.59)	211.04 (54.87)	13.743 (.459)	9.391 (.533)
Age squared	-21.45 (.75)	-12.62 (2.21)	-15.12 (.80)	-4.79 (1.89)	-0.385 (.015)	-0.252 (.018)
Education	56.69 (6.41)	14.22 (17.07)	114.90 (6.14)	-56.03 (15.03)	1.262 (.132)	3.086 (.139)
Education squared	-1.58 (.25)	-0.21 (1.22)	-2.22 (.24)	5.97 (1.05)	-0.041 (.005)	-0.068 (.006)
Unemployment rate	-9.98 (3.85)	-31.37 (14.58)	-15.94 (4.15)	-42.24 (13.00)	-0.130 (.079)	-0.304 (.094)
Unemployment rate × kids	5.27 (4.17)	33.60 (13.44)	1.33 (4.14)	34.40 (11.10)	0.054 (.086)	-.065 (.094)
Maximum monthly AFDC benefit	-0.22 (.06)	-0.10 (.18)	-0.54 (.06)	-0.14 (.14)	-0.005 (.001)	-.014 (.001)
Kids (γ_0)	-83.03 (47.82)	-249.44 (132.61)	-186.48 (46.65)	-327.07 (110.24)	-6.856 (.981)	-11.420 (1.054)
Post86 (γ_1)	-29.95 (23.61)	63.27 (78.03)	-45.33 (25.20)	-56.27 (69.26)	0.722 (.484)	0.222 (.569)
Kids × Post86 (γ_2)	25.22 (15.18)	2.98 (46.04)	37.37 (15.31)	83.83 (39.42)	.126 (.311)	.560 (.346)
Observations	59,474	5700	67,097	9354	59,474	67,097

Key findings and discussion

Other key findings

- Increase in labor supply of treated group not explained by different long-run trends
- Neither do other changes in late 80s (Other programs & state-wide unemployment)
- Large participation increase but minimal labor hour changes
 - ▶ Pretty common in the literature: Work hour usually fixed by contract

Discussion on the overall EITC impacts

- Labor supply changes are only part of the evaluation
- How much labor supply changes/distortion exists *compared to other programs* matters
- What people do with the extra money matters: Expenditure, savings, assets etc

Kleven (2024): The EITC and the Extensive Margin: A Reappraisal

Q: Are the previous studies on EITC correct on labor supply effects?

Shifts in assisting the poor to the *working* poor

- Number of recipients of EITC = 30 times the number of other welfare
- Expansion of EITC based on observation that labor supply at extensive margin $\uparrow$
- Traditional welfare, on the other hand, blamed for labor disincentives

This paper reassess previous labor supply evidence on EITC

- Long-run analysis of all EITC reforms since 1975
- Employs event-study approach: Fuller characterization of dynamic effects
- Tests previous findings with different empirical models
- Distinguishes pure EITC effects from confounding factors
- While focusing on the same population: Single women with and without children

Data on EITC reforms studied in this period

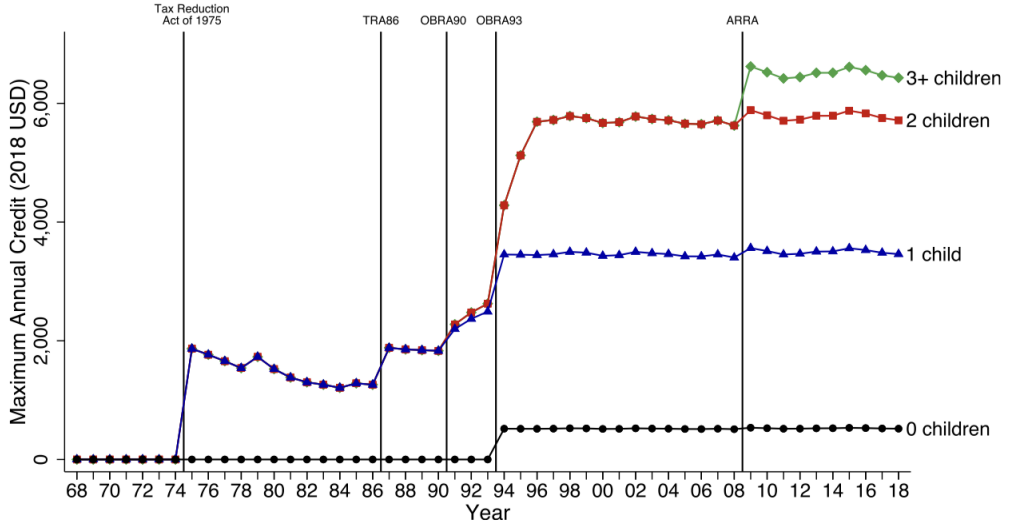
EITC and welfare reforms

- Federal: '75, '86 (studied in previous paper), '90, '93, '09
 - ▶ '75 (EITC introduction) is the largest, followed by '93 expansion
- State: 30 states introduced EITC supplements in '84-'18
- Others: Various state-level reforms culminated to nationwide-level at '92-'96
 - ▶ Introduction of Temporary Assistance for Needy Families (Families)
 - ▶ Time limits and work requirements, but benefits strictly increasing in children

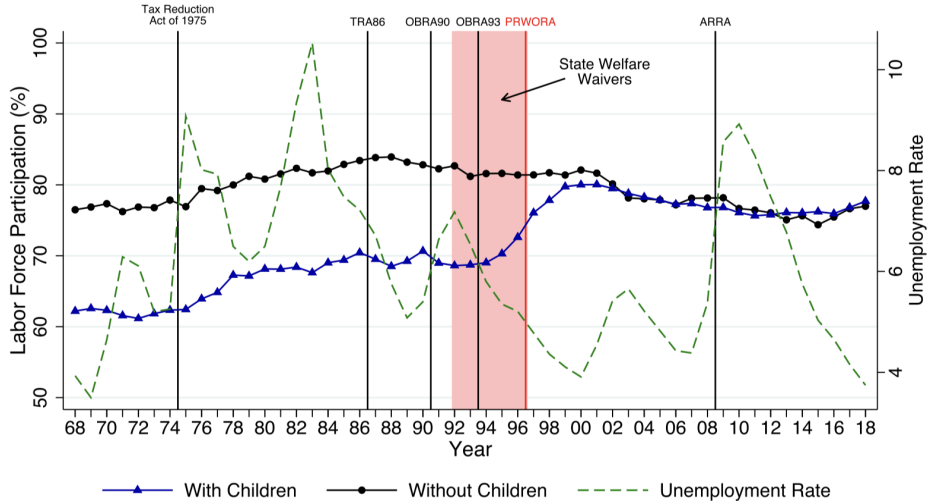
Data and empirical strategy

- Demographics and socioeconomic status gathered from Current Population Survey
- Treatment group will be single women with children
- But also analyze heterogeneity of EITC effects across following categories
 - ▶ Heterogeneity by predicted earnings based on observables
 - ▶ Education, # of children, age of youngest children, race, and state

Changes in EITC setup over time (Max credit)

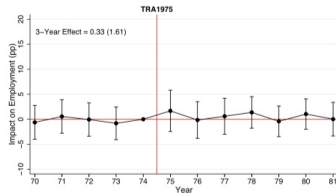


Limited labor supply changes except for '92-'96

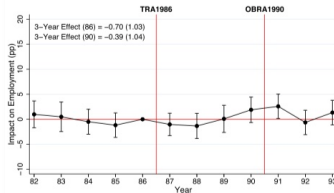


...also the case when event studies are applied

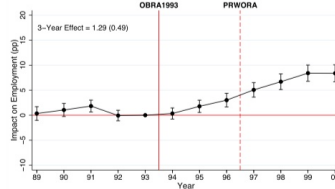
A: 1975 Reform



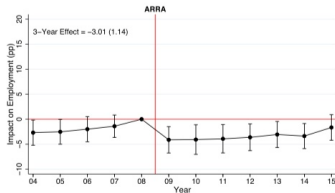
B: 1986 & 1990 Reforms



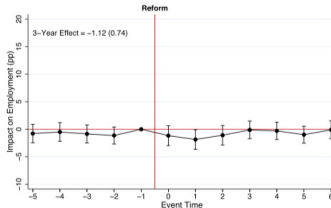
C: 1993 Reform



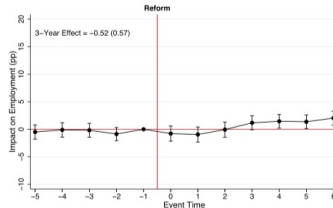
D: 2009 Reform



E: Federal Reforms Stacked, Without 1993

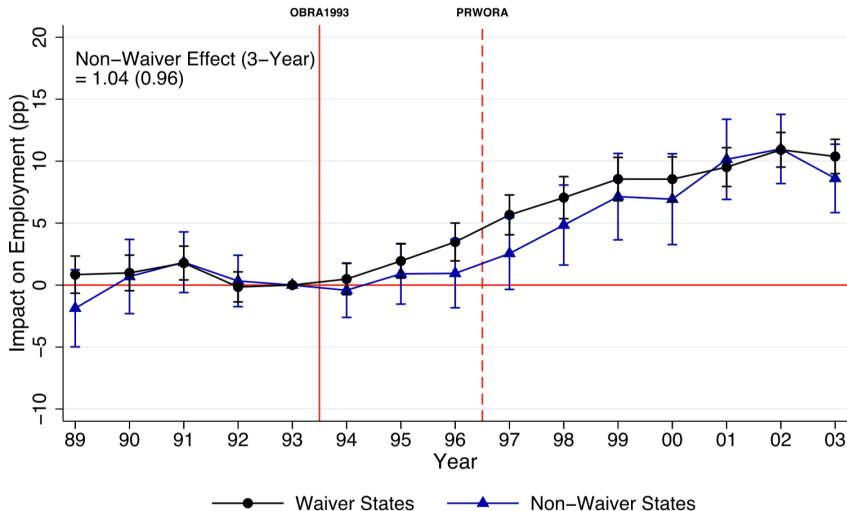


F: Federal Reforms Stacked, With 1993



Note: This pattern holds when only those with bottom half of predicted earnings are included (Fig 6 in text)

Effects mostly driven by state-level waiver reforms



Net out waiver reform states and control for macroeconomy (unemployment) → Event-study for '96 reform?

Other findings and discussion

So what does this imply about findings on EITC?

- Many econometric specifications only capture short-run effects
- Treatment status based on fertility may not be the best approach
- Ignoring contemporaneous effects may mask effects of potential confounders

So does this mean that EITC is ineffective? Not necessarily

- Effects on other outcomes (health, household finances) still studied
- Ultimately depends on redistribution
 - ▶ Is it financed by those who are able to finance them?
 - ▶ Are working poor receiving higher consideration compared to the average?
- Optimality depends on holistic set of goals, not just labor supply responses

Unanswered questions

Overall effect on reducing extreme poverty

- Conflicting finding on labor supply, a proxy for long-run self-sustenance
- Many household finance outcomes still being explored
- Implementation of graduation programs and studies on their effects underway
 - ▶ Time-limited, structured programs to get ultra-poor out of poverty
 - ▶ Bundle transfers with job training, asset accumulation, household finance
 - ▶ Aim for eventual self-sustenance and decreased reliance on welfare programs

Room for further research on other aspects

- How much does implementation (not program design) matter?
 - ▶ Many incidences of incapable and/or corruptible agents
 - ▶ How much of the (lack of) effect is attributable to poor implementation vs poor design?
 - ▶ Can implementation of technology reduce administrative burden, easing implementation?
- How to implement such programs while minimizing political distaste?